

Brainlift: MCAT Preparation

Owner

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Purpose

This Brainlift focuses on the dynamics of MCAT preparation, including studying tools and methods that help diverse groups of people use to prepare for the MCAT. It focuses on Anki, a flashcard app that uses spaced repetition and active recall to optimize learning.

In Scope

- Discuss the various MCAT study methods and their outcomes
- Analyze MCAT applications and their pros and cons
- Identify gaps in MCAT preparation techniques and attempt to find solutions

Out of Scope

- Too in depth into the content on the MCAT
- College classes that are useful for the MCAT

DOK 4: Spiky Points of View (SPOVs)

Spiky POV 1: A good all-in-one MCAT study tool can and should exist.

Elaboration: I'll split this into two.

- **An all-in-one MCAT study tool should exist.** This is not too controversial, unless you are one of the big companies that profits from making premed students pay thousands of dollars on different resources and spend hundreds of hours stressing about which ones to pay for. The fact that so many different resources exist and not one of them is sufficient is not by fault, it's by design. The truth is, it would be hard to make a comprehensive MCAT study tool. As you'll see in my other DOKs, the MCAT study plan itself is split into 3 main phases: understanding, memorizing, and applying. Combining these is difficult because of how much ground each of them covers, and how each has unique structural requirements. But another reason is because of the massive monopolies like UWorld spending millions of dollars funding medical writers specifically to craft hyper-detailed, visual explanations, and AAMC literally being the writers of the test and barring others from copying past exam questions. Trying to do all spreads your resources too thin, unless you do it right. An all-in-one MCAT app would prevent students from needing to keep track of which concepts they need to review, their study schedule, what other resources they have

to buy, how to combine their different score predictions from different sources, and having to adjust each resource when the state of another changes.

- **An all-in-one MCAT study tool can exist.** By taking the best from each app, I believe that combining features doesn't spread my resources too thin, but actually synergizes them well to make a better product. The fact is, many people already combine separate apps in specific ways to optimize their study routines. For example, one popular study tip website suggests that users "Do UWorld first, then add weak points to Anki. This prevents false confidence and reveals what you actually need to review" (Category 2, Source 5). Why can't we just directly integrate UWorld's questions into Anki, have Anki analyze strengths and weaknesses, and then adjust flashcard recommendations based on that information? There are so many apps that are cited as "complementing" each other, but the fact that they complement each other means that they can be combined and integrated not only to consolidate the different sections, but also to have each section enhance the others. By taking advantage of the information each section provides we can create reliable, data-driven predictions and suggestions that will provide clarity and direction to learners while also saving them lots of time from not having to deal with a million different apps in their study schedule.

The fact that premed students have to spend time researching and purchasing multiple expensive tools and keep track of when to do each of them is a clear signal that the current system is broken. Consolidation will simplify the process by organizing various resources into one simple system while combining information to create strong predictions and recommendations. **I can't do this in one week**, but I think that I can create the kernel for predictions and recommendations by having my app create a schedule by ingesting practice tests and re-prioritizing review by topic weakness.

Spiky POV 2: MCAT study tools aren't using AI correctly.

Elaboration: Even though many tools are using AI to generate questions, explain problems, plan study schedules, and assess mastery across various topics, I don't think they're using AI to their full potential. AI right now has immense potential, especially when it comes to generation, but only if done right with systems like RAG that ground the AI in truth from external sources. I think AI can be used to create visuals in questions and lessons to go in depth into concepts; AI voice assistants should also be used to quiz students live. I think implementing multi-modal AI into MCAT study tools helps auditory and visual learners as well as other types of learners. Additionally, AI can be used to sift through practice questions, flashcards, review sessions, and identify weaknesses and strengths. Based on those, it could recommend users focus on certain concepts, prioritize different types of questions, and be used to having each different learning phase build on each other. A major problem with MCAT studying is the disconnected tools and resources needed to build expertise on various different subjects and types of questions. Due to the disconnection, users may focus on the wrong subjects, focus too much on one type of resource, or spend lots of time gathering all the information about their current mastery levels. AI can mitigate these problems and save lots of time and energy so that MCAT testers can focus on learning.

Experts

Expert 1: Dr. Ken Tao

Who: Dr. Ken Tao, PhD in Neuroscience (Harvard), VP of Product Development & Innovation at MedSchoolCoach

Focus: Renowned for his dual expertise in MCAT preparation and mentoring students in research, he leads a top-tier tutoring team and has an exceptional track record in driving student score improvements.

Why Follow: Nationally recognized as an exceptional MCAT mentor, Dr. Tao has over 10,000 hours of teaching experience through roles as a graduate teaching assistant, adjunct faculty member, and MCAT instructor/tutor at the Princeton Review and MedSchoolCoach. At the Princeton Review, he was the only tutor certified in all MCAT subjects, one of their highest-rated tutors, and a teacher trainer for their MCAT division. His contributions include authoring content for numerous books and guides; many of his students have achieved scores of 520+ on the current exam.

Where: LinkedIn: [linkedin.com/in/ken-tao-83510027](https://www.linkedin.com/in/ken-tao-83510027) | YouTube: @MedSchoolCoachMCATPrep

Expert 2: Jack Westin

Who: Jack Westin, Founder of Jack Westin MCAT Prep

Focus: MCAT CARS mastery and comprehensive full-course content review. Known for producing high-quality free daily CARS passages and targeted, exam-specific content videos.

Why Follow: Jack Westin's platform is one of the most widely cited free MCAT resources in the r/MCAT community. His course is specifically designed for students with weaker science backgrounds, teaching only what the exam tests and cutting the noise found in overly broad prep books. His CARS practice passages and video library indexed by topic are staples in successful study plans referenced throughout this BrainLift.

Where: jackwestin.com | YouTube: Jack Westin | Reddit: r/MCAT

Expert 3: Kevin Jubbal, MD

Who: Kevin Jubbal, MD, Founder of Med School Insiders

Focus: Evidence-based study strategies, productivity systems, and pre-med optimization. Combines cognitive science research with practical guidance for MCAT prep and medical school success.

Why Follow: Dr. Jubbal bridges the gap between academic research on learning—spaced repetition, active recall, interleaving—and practical MCAT application. His frameworks for building efficient study schedules, diagnosing knowledge gaps, and avoiding common pitfalls align directly with the core thesis of this BrainLift: that deliberate, high-yield, understanding-first studying beats passive review and volume-chasing every time.

Where: medschoolinsiders.com | YouTube: Med School Insiders | Instagram: @medschoolinsiders

DOK 3: Insights

Insight 1: Studying for the MCAT is incredibly fragmented.

Learning the MCAT has 3 main phases: understanding, memorizing, and applying. **There is no app that effectively consolidates all 3 of these phases effectively.** Anki is the best for memorizing but not for understanding or applying. Practice tests and UWorld are good for applying knowledge, Khan Academy is good for understanding. Jack Westin is good at teaching understanding and applying, but it is not optimized for memorization. Additionally, students often have to keep track of error logs, concept mastery sheets, study schedules, and past practice tests. If this could all be integrated into one app, it could save massive time for test-takers and help them focus on studying rather than organizing.

Insight 2: Different people require different preparation methods.

The MCAT is taken by various different people, from students with little exposure to science to students who have taken various undergraduate courses in science concepts. Additionally, different people have different schedules, different endurance levels when it comes to studying, different budgets, and different goals. With all of these different variables to consider, creating the correct preparation method can be difficult, and sticking to it can be even harder. If an app could consider all these different variables, craft a personalized study plan for users, and help them maintain it, it could alleviate many burdens from the user and greatly increase their studying efficiency. AI could be especially useful for personalization.

Insight 3: AI has been used to significantly improve MCAT preparation

Numerous MCAT apps today use AI to optimize different parts of the study sessions. Anki's spaced repetition algorithm uses the machine-learning FSRS to more efficiently space card intervals, leading to less time for the same learning. AI is used by numerous apps to give explanations to questions, generate problems, act as a guide for students when tutors aren't available, and create adaptive study plans based on user's preferences and mastery levels. All of these processes save time for tutors and question creators while allowing learners to instantly create tests, receive feedback, ask questions, and go in depth into various different MCAT topics. This personalization is incredibly useful for users and is also cheap and scalable when compared to the busy work schedules of tutors.

DOK 2: Knowledge Tree

The Knowledge Tree is the structured foundation of the BrainLift (Depth of Knowledge Level 2). It organizes information from expert sources in a hierarchical system, from broad categories to specific facts.

Category 1: Anki

Subcategory 1.1: Anki Overview & Features

Source 1: Anki Official Website

DOK 1 - Facts:

- Anki is a flashcard program that helps you spend more time on challenging material, and less on what you already know.
- The free AnkiWeb synchronization service lets you sync your cards across devices.
- Audio, images, videos, and scientific markup can be used in your flashcards.
- Easily change your flashcard layouts and the timing of their reviews.
- Anki can handle decks of 100,000+ cards with no problems.
- Install add-ons to extend Anki with new features and functionality.

DOK 2 - Summary:

Anki is a flashcard program that focuses on building on your weaknesses while maintaining your strengths through efficiently timed learning (spaced repetition). It allows you to sync your cards across devices, and these cards can have audio, images, videos, and scientific symbols. It can handle 100,000+ card decks, and has many add-ons to extend features and functionality.

Link to source: <https://apps.ankiweb.net>

Source 2: Anki Official FAQ

DOK 1 - Facts:

- Anki uses two algorithms: SM-2 (SuperMemo 2) and FSRS.
- SM-2
 - You control the initial learning steps (not fixed at 1 then 6 days)
 - 4 answer choices instead of 6; only one "fail" option
 - Late reviews get a boost factored into the next interval
 - Failed cards can have interval reduced rather than fully reset
 - "Easy" adds an extra bonus on top of the ease factor increase
 - Repeated failures during learning don't lower a card's ease (avoids "low interval hell")
- FSRS
 - Retrievability (R): probability of recalling a card right now
 - Stability (S): days until R drops from 100% to 90%

- Difficulty (D): how hard it is to increase stability after review
- Each card has its own DSR "memory state." FSRS uses machine learning on your review history to fit parameters automatically — don't tweak them manually. Instead, set your desired retention level. Higher retention = more reviews.
- FSRS advantages over SM-2: fewer reviews for the same retention, and much better handling of cards reviewed after a long break.

DOK 2 - Summary:

Anki uses 2 algorithms for its spaced repetition algorithm, SM-2, which adjusts the card's interval and ease factor based on your previous answers, and FSRS, which uses machine-learning algorithms to measure Retrievability, Stability, and Difficulty to calculate intervals. FSRS is overall better, allowing for fewer reviews for the same retention, and much better handling of cards reviewed after a long break.

Link to source: <https://faqs.ankiweb.net/what-spaced-repetition-algorithm>

Subcategory 1.2: Research & Academic Evidence

Source 3: Michael A. Nielsen, "Augmenting Long-term Memory"

DOK 1 - Facts:

- The single biggest change that Anki brings about is that it means memory is no longer a haphazard event, to be left to chance. Rather, it guarantees I will remember something, with minimal effort. That is, Anki makes memory a choice.
- What makes Anki better than conventional flashcards is that it manages the review schedule. If you can answer a question correctly, the time interval between reviews gradually expands. So a one-day gap between reviews becomes two days, then six days, then a fortnight, and so on. The idea is that the information is becoming more firmly embedded in your memory, and so requires less frequent review. But if you ever miss an answer, the schedule resets, and you again have to build up the time interval between reviews.
- While it's obviously useful that the computer manages the interval between reviews, it perhaps doesn't seem like that big a deal. The punchline is that this turns out to be a far more efficient way to remember information.
- Anki saves time by a factor of more than 20 compared to conventional flashcards, because they know how to manipulate spaced repetition to save time and focus on concepts that are particularly weak for the user.

DOK 2 - Summary:

Anki saves lots of time reviewing concepts by optimizing review schedules.

Link to source: <https://augmentingcognition.com/ltn.html>

Source 4: A Cohort Study Assessing the Impact of Anki as a Spaced Repetition Tool on Academic Performance in Medical School

DOK 1 - Facts:

- Anki is an application that capitalizes upon the techniques of spaced repetition and is increasingly utilized by medical students for examination preparation.
- Anki users scored significantly higher across all four exams: Course I (6.4%; $p < 0.001$); Course II (6.2%; $p = 0.002$); Course III (7.0%; $p = 0.002$); and CBSE (12.9%; $p = 0.003$). Students who reported higher dependency on Anki performed significantly better.
- Anki is an open-sourced flashcard system with feedback-rich statistics and many community-developed decks. The AnKing deck, with over 30,000 flashcards, was the most utilized deck by medical students.
- Course II, which requires memorization of new vocabulary (e.g., microorganisms, antibiotics), is well-suited to the flashcard + spaced repetition + retrieval-practice combination.
- Course III performance had the lowest correlation with daily average Anki cards; this module emphasizes integration of principles (cardiology, pulmonology, nephrology) over memorization.
- Anki modalities emphasize rote learning and recall. Applying knowledge differs because it focuses on using acquired information to solve problems, make decisions, or perform tasks.
- Learning through Anki may not always facilitate transfer of knowledge to new or unfamiliar situations.

DOK 2 - Summary:

Anki is very good for rote learning and is particularly useful for such concepts, leading to significant score increases. For more application based learning, it is less useful. It is often not useful for unfamiliar questions .

Link to source: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10403443/>

Subcategory 1.3: MCAT-Specific Anki Strategies

Source 6: How to Use Anki for the MCAT in 2026

DOK 1 - Facts:

- Anki should never be your first exposure to new material. Learn the material first using textbooks, videos, or lectures, then use Anki to reinforce and retain. Anki strengthens knowledge. It does not create it.
- Doing Anki inconsistently is the fastest way to make it feel overwhelming. Skipped days cause reviews to pile up. Doing fewer cards every day beats doing hundreds sporadically. Anki rewards consistency, not heroics.

- Reviews are where memory is built. New cards create future workload.
- More cards does not mean more learning. Most students hurt themselves by trying to complete every deck, every tag, or every card available.
- Never skip reviews.
- Recommended: 100 to 150 new cards per day; complete all scheduled reviews every day.
- Use one main deck. Add supplementary decks only to target weak areas, most commonly Psych/Soc.
- The AnKing Heatmap visually tracks your daily review streak. It makes missed days obvious, encourages daily engagement even on low-energy days, and reinforces discipline without adding cognitive load.
- AnkiBrain can help explain cards you consistently miss by providing quick context or reworded explanations. It should be used sparingly—it is a clarification tool, not a replacement for real studying.

DOK 2 - Summary:

Anki shouldn't be used for first exposure to material (learning), but to review on top of learning foundations. New cards and reviews need to be done consistently. More cards doesn't mean more learning; use one main deck, add supplementary decks to target weak areas. Add extensions like AnKing Heatmap and AnkiBrain for keeping yourself consistent and learning.

Link to source:

<https://thetatchguy.com/how-to-use-anki-for-mcat-best-decks-study-schedule/>

Category 2: Studying for the MCAT in General

Subcategory 2.1: Core Study Strategies

Source 1: Studying Strategies (MedSchoolInsiders)

DOK 1 - Facts:

- Doing well on the MCAT comes down to three main factors: Understanding, Memorizing, and Applying.
- The quality of spaced repetition resources is critically important—not just frequency of use. You must have proper intensity, volume, and form.
- There is an upper limit on the amount of information that can be memorized in a given period. Focus on high-yield concepts most likely to impact your score. Deprioritize low-yield information because of the forgetting curve.
- Through Anki, students are often encouraged to focus on memorizing without the critical preceding step of understanding. Without a foundation of comprehension, it is far less effective to memorize facts in isolation.

DOK 2 - Summary:

These are the 3 steps of learning the MCAT: understanding, memorizing, and applying. Spaced repetition needs to be done right; you need the right intensity and volume to maximize learning, and too much information can be bad. Before doing Anki students need to gain an understanding.

Link to source: <https://medschoolinsiders.com/study-strategies/fix-your-mcat-studying/>

Subcategory 2.2: Study Planning & Scheduling

Source 2: 5-Month Study Plan

DOK 1 - Facts:

- The five-month sweet spot allows approximately two weeks per content chapter with built-in review time.
- Anki flashcard decks, particularly MileDown or JackSparrow, appear in nearly every successful study plan. Reddit data shows students using Anki daily score an average of 3 points higher.
- High-scoring students used: 1 hour content review in the morning, 30 minutes of Anki during lunch, and 1 hour of practice problems in the evening. Weekends were for full-length tests every other Saturday and review on Sundays.
- Students who tracked their hours weekly and adjusted based on performance were 40% more likely to reach their target scores. Consistency over intensity—moderate daily studying outperforms irregular marathon sessions.
- Regular self-assessment is essential. One Reddit user found 60% of their biology errors came from just three content areas: enzymes, the immune system, and genetics.
- The "review every question method": after each practice test, spend 2-3 hours reviewing not just wrong answers but also questions you guessed on correctly.
- Reddit users recommend 8-10 full-length practice tests during a 5-month period, spaced approximately every two weeks during the final three months.
- Create an error log categorizing mistakes as content gaps, timing issues, or misreading questions.
- The final two weeks should focus on reinforcing knowledge rather than learning new material.
- The P/S section particularly benefits from last-minute review—it is the most "crammable" section.

DOK 2 - Summary:

Five months is a good amount of time to study, giving two weeks per content chapter. Anki allows for 3 points extra. Reviews should have a consistent daily schedule (1 hour morning 1 hour evening, review in the middle). Practice tests should be taken regularly.

Tracking hours leads and adjusting leads to higher scores. You should take 8-10 full length practice tests spaced somewhat evenly (biased towards the final 3 months). You should review every single question from your practice tests. Error logs categorizing mistakes as content gaps, timing issues, or misreads are useful. The final two weeks are for reinforcement, not new content. The P/S section is crammable.

Link to source:

<https://atutor.ca/why-reddits-5-month-mcat-study-plan-actually-works-when-others-fail/>

Subcategory 2.3: Resources & Tools

Source 3: Free Resources

DOK 1 - Facts:

- Khan Academy is one of the best resources for gaining initial comprehension.
- Anki is good for review.
- Jack Westin has lots of review guides and test preparation materials.
- AAMC questions are necessary for testing and applying.

DOK 2 - Summary:

Khan Academy is the best free resource for learning, Anki and Jack Westin are good for review, and AAMC is good for applying and testing.

Link to source:

<https://arvindrajan.notion.site/The-Ultimate-MCAT-Free-Resource-Compilation-fcff61a7f99a4f13871dde51ca5cf4ab>

Source 4: Jack Westin (Reddit Review)

DOK 1 - Facts:

- Jack Westin offers a large number of free resources.
- The course is recommended for people with a limited science background or who are weak in orgo/biochem—it teaches only what is required on the exam.
- Many prep books have an entire section on orgo when realistically a 4-5 hour lecture could cover most of what is tested.
- Jack Westin's content lecture videos are indexed by topic, allowing targeted re-watching of high-yield material.

DOK 2 - Summary:

Jack Westin is good for learning with a limited science background. Jack Westin's content lecture videos being indexed by topic is useful for re-watching important material.

Link to source:

https://www.reddit.com/r/Mcat/comments/x9avlj/honest_review_jack_westin_full_course_review_502/

Source 5: UWorld**DOK 1 - Facts:**

- UWorld stands out for two reasons: exceptional explanations that deepen understanding of concepts and reasoning, and challenging questions that strengthen the cognitive skills required for the MCAT.
- UWorld forces engagement with unfamiliar passages, interpreting graphs, integrating content, and justifying choices.
- Priority order for resources: AAMC > Full-length exams > Section Bank > UWorld > Optional restart.
- Do Anki before UWorld to "warm up" recall pathways and improve confidence entering a block.
- Do UWorld first, then add weak points to Anki. This prevents false confidence and reveals what you actually need to review.
- Final 6 weeks: Use AAMC exclusively.
- UWorld tends to be more detailed, more content heavy, more graph-intensive, and more verbose than the actual MCAT.
- UWorld is for learning, not scoring. It is very hard to draw reliable comparisons between UWorld and your MCAT score.

DOK 2 - Summary:

UWorld has challenging questions with good explanations which are useful for learning and applying skills. By forcing engagement with unfamiliar media it provides good practice for those hard passage-based MCAT questions. Anki can be done before UWorld for brief refreshers and after to review mistakes. UWorld tends to be more detailed, more content heavy, more graph-intensive, and more verbose than the actual MCAT.

Link to source: <https://thematchguy.com/how-to-use-uworld-for-mcat/>

Source 5: Roediger & Karpicke (2006) — The Testing Effect**DOK 1 - Facts:**

- Active recall is a learning technique where you test yourself on information rather than passively rereading it; spaced repetition is reviewing that information at gradually increasing intervals. Together, they form the most effective combination for long-term memory retention that cognitive science has identified.
- In a landmark study, students who used retrieval practice retained approximately 80% of the material after one week, compared to just 34% for those who only reread. A single act of retrieval can more than double long-term retention.
- Practice testing (active recall) and distributed practice (spaced repetition) were the only two techniques rated as "high utility" in a comprehensive review of ten common

learning techniques (Dunlosky et al., 2013). Popular methods like highlighting, rereading, and summarization were rated "low utility."

- Retrieval practice produces 50% more correct responses on subsequent tests than elaborative studying with concept mapping (Karpicke & Blunt, 2011).
- Neuroimaging shows that active recall activates the hippocampus and prefrontal cortex together, strengthening neural connections. Repeated retrieval gradually shifts memories from hippocampal to neocortical storage, making them more stable and resistant to interference.
- Without any review, most people forget roughly 50% of new information within an hour and up to 70% within 24 hours (Ebbinghaus forgetting curve).
- Each time material is reviewed, the forgetting curve flattens and intervals between required reviews grow longer.

DOK 2 - Summary:

The 2 most effective study techniques are practice testing (active recall) and distributed practice (space repetition). Retrieval practice leads to 80% retention as opposed to rereading which is 34%. Retrieval practice produces 50% more correct answers; active recall strengthens neural connections by activating different parts of the brain simultaneously. Without review, people forgot 50% of new information in an hour, 70% in 24 hours. The forgetting curve flattens as material is reviewed, greater intervals between next required reviews.

Link to source: <https://thematchguy.com/how-to-use-uworld-for-mcat/>

Subcategory 2.4: Memory Science & Learning Theory

Source 6: The Science of Memory — Evidence-Based Study Strategies for MCAT Success

DOK 1 - Facts:

- You may understand a concept in the moment (encoding), but being able to pull it out under timed test conditions (retrieval) requires practice.
- Spaced repetition: revisiting material at increasing intervals over time reduces the forgetting curve and strengthens long-term memory.
- The effort of retrieval strengthens memory—don't passively flip cards; try to recall the answer before checking it.
- Active Recall means actively retrieving information from memory rather than passively reviewing notes.
- Active recall methods: convert notes into Q&A format; practice with AAMC question banks and timed sections; immediately quiz yourself after reviewing content without looking at answers.

DOK 2 - Summary:

Retrieval requires practice in timed conditions. Space repetition helps forge long-term memory. Students should recall the answer before checking it. Reviewing notes is passive. You should convert notes into Q&A format; practice with AAMC question banks and timed sections; immediately quiz yourself after reviewing content without looking at answers.

Link to source: <https://thematchguy.com/evidence-based-mcat-study-strategies-memory/>

Source 7: Khan Academy — AAMC Partnership & Structure

DOK 1 - Facts:

- Khan Academy's MCAT prep course is the first and only to be officially approved by the AAMC, the administrators of the MCAT. Its structure follows AAMC Foundational Concepts rather than being organized by subject.
- Khan Academy covers all four MCAT sections with in-depth instructional videos and approximately 20 practice passages per section. It is entirely free with no time limit.
- Khan Academy's Psychology/Sociology section is particularly well-regarded and frequently cited as essential prep for the P/S section.
- Khan Academy is best suited to visual and auditory learners. It is stronger for initial comprehension than for active problem-solving practice. Students who prefer textbook-style learning are better served by resources like Kaplan or ExamCrackers.
- Khan Academy lacks a spaced repetition system and does not adaptively schedule reviews, meaning students must self-manage when and how they return to material.

DOK 2 - Summary:

Khan Academy is good at introducing students to the concepts and helping them to understand them, but may not be suited for memorization and application problems. It's good for visual and auditory learners. Studying must be self-managed by students.

Link to source:

<https://elitemedicalprep.com/khan-academy-for-the-mcat-professional-review/>

Subcategory 2.4: AI and the MCAT

Source 8: How to Use AI to Study for the MCAT

DOK 1 - Facts:

- AI can analyze your performance on practice tests to identify your strengths and weaknesses.
- AI can analyze your performance on practice tests to identify your strengths and weaknesses.

- Based on your learning style (visual, auditory, kinesthetic), AI can suggest the best resources.
- Platforms like Kaplan use AI to help write explanations and adjust question difficulty based on your performance. If you're excelling, you'll get harder questions; if you're struggling, you'll get easier ones to reinforce basic concepts.
- You can also pair this with your MCAT Self Prep end of lesson quiz questions by using generative AI like ChatGPT or Grok to create similar MCAT-styled questions based on the same topics.
- AI can identify patterns in the types of questions you answer incorrectly, helping you focus on those concepts.
- AI-powered tutoring bots or generative AI like ChatGPT or Grok can offer explanations, answer questions, and help to clarify doubts at any hour, ensuring you're never stuck.
- Natural Language Processing (NLP) tools can summarize complex texts from your MCAT books or lectures, making dense information more digestible.
- AI can help you plan your study sessions to maximize efficiency.
- AI tools provide dashboards showing your study progress, helping you to see how much you've covered and what remains.
- AI can simulate the exact conditions of the MCAT, from question types to time constraints, using platforms like Blueprint MCAT.
- After taking practice tests, AI can predict your score range, allowing you to adjust your preparation strategy accordingly.

DOK 2 - Summary:

AI is very good at personalization; for example, it can be used to analyze your performance to suggest which concepts to review and give you a good study schedule based on your preferences. It is good at creating questions when given good context, explaining right and wrong answers, summarizing large media, showing your progress, and predicting your success.

Link to source: <https://mcatselfprep.com/how-to-use-ai-to-study-for-the-mcat/>

Category 3: What is the MCAT even about

Source 1: What's on the MCAT® Exam?

DOK 1 - Facts:

- The Biological and Biochemical Foundations of Living Systems section asks you to solve problems by combining your knowledge of biological and biochemical concepts with your scientific inquiry and reasoning skills.
- The Psychological, Social, and Biological Foundations of Behavior section asks you to solve problems by combining your knowledge of foundational concepts with your scientific inquiry and reasoning skills.
- The Chemical and Physical Foundations of Biological Systems section asks you to solve problems by combining your knowledge of chemical and physical foundational concepts with your scientific inquiry and reasoning skills.
- The Critical Analysis and Reasoning Skills section of the MCAT exam will be similar to many of the verbal reasoning tests you have taken in your academic career. It includes passages and questions that test your ability to comprehend what you read.

DOK 2 - Summary:

The MCAT has biology, biochemistry, psychology, social and biological behavioral science, chemistry, physics, and critical analysis. These are split into 4 sections: The Biological and Biochemical Foundations of Living Systems, The Psychological, Social, and Biological Foundations of Behavior, The Chemical and Physical Foundations of Biological Systems, and The Critical Analysis and Reasoning Skills.

Link to source:

<https://students-residents.aamc.org/prepare-mcat-exam/whats-mcat-exam-pdf-outline>

Source 2: MCAT Statistics

DOK 1 - Facts:

- Popular test pre websites suggest students invest 300–350 hours in MCAT prep.
- Live online prep courses cost \$2,000–\$6,500 on average. Classroom courses range from \$2,500–\$7,000+. MCAT tutors charge \$50–\$150/hour.

DOK 2 - Summary:

Studying for the MCAT is expensive, time wise and money wise.

Link to source: <https://tutors.com/costs/mcat-cost>

<https://www.kaptest.com/study/mcat/whats-a-good-mcat-score/>

<https://testpreppal.com/mcat/tools/mcat-statistics>